

D7: Healthcare Technology Design Project – Part 1

Module: BIOM2011 & 2012

Lecturer: Dr Ernesto E. Vidal-Rosas

Deadline of Report: 22nd April 2026

Demonstration: 9th March 2026

Weighting: 10% of the module

Relevant Learning Outcomes (LOs)

1. Analysis of a real-world healthcare technology design problem from its functional requirement point of view
2. To construct different modules in a multidisciplinary problem and integrate them in a complete system
3. Understanding rigorous validation of a complex system from the functional requirements point of view
4. Learning hardware design techniques for a biomedical system
5. To analyse the results and write a scientific report

This is a design project. In your 1st year and in the 1st semester this year you have learned the basics required for this design project from several labs where the lab instructions told you what exactly you were supposed to do. This is different here – you have to apply your knowledge and do research to implement the design.

Marking scheme

<i>Criterion</i>	<i>Description</i>	<i>LOs</i>	<i>Total</i>
1	Development of ECG pre-processing filter in Matlab	2	20%
2	Mapping the filter to an efficient hardware architecture	1-5	30%
3	Synthesis of hardware on FPGA	3-5	30%
4	Scientific report writing	5	10%

Evaluation Method

1. Individual understanding through a viva during demonstration (covering criteria 1, 2 and 3)
2. Report writing (covering criterion 4)

1. Aims

To design a hardware filter for recovering clean ECG signal from a corrupted signal and analysis of performance of the hardware.

2. Mode of working

This is an individual project and will run every Monday from 29th January until 12th February. You have 18 hours lab time scheduled for it. The work is divided in three steps: 1) algorithmic design of a filter for removing noise and artefacts of the given ECG signal using Matlab; 2) design of the filter using SystemVerilog HDL and its functional validation using Modelsim; 3) synthesis and implementation of the filter on a DE1-SoC FPGA development board and its validation; 4) write a scientific paper (IEEE 2-columns format, 4-page length).

3. The problem

This exercise is, in one sense, a continuation of the lab work you have done as a part of ELEC 1212 module in your 1st year.

ECG is an extremely important signal for understanding the state of heart function of a person. The ECG signal is captured by dedicated electrodes placed on the chest area. However, this signal is corrupted with noise and artefacts. To analyse the ECG, therefore, the first step is to remove these noise and artefacts – a step called Pre-processing of ECG. Filters are automatic choice for such processing as they can selectively remove frequency components (or bands) in a signal.

With the prominence of wearable devices – ECG is one of them – it is necessary to design and implement algorithms in such a way that the battery of the system lasts at least for 16 – 18 hours. For that, one may need to design the pre-processing architecture in hardware as hardware solutions are more power efficient than a software solution.

In this exercise you will develop a hardware filter for pre-processing ECG going through the entire chain of VLSI design process – algorithm

development (Matlab), architectural mapping of the algorithm (SystemVerilog structural modelling) and its hardware implementation (FPGA). Each of these steps will involve validation and performance analysis of the system. The filter should run at least at a frequency of 10 KHz.

In terms of filters, you have two choices, 1) Finite Impulse Response (FIR), and 2) Infinite Impulse Response (IIR). FIR filter design is straightforward, but it may need more hardware. On the other hand, IIR filter will require less hardware but because of some feedback loop it might get unstable.

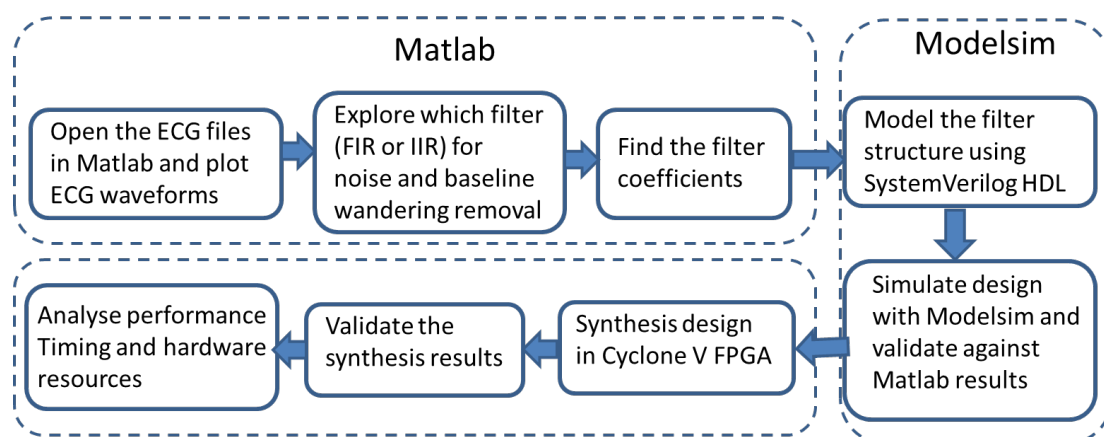
4. Preparation

In different folders you have all the resources you will need:

1. Folder ECG: contains 4 ECG signals that you need to open in Matlab – the signals are sampled at 1 KHz and contains both noise and baseline wandering artefacts.
2. Folder Tutorial: this folder contains two tutorials on how to simulate a SystemVerilog design in Modelsim and how to do synthesis of the SystemVerilog code in Quartus FPGA.
3. Folder Resource: contains materials on different types of adders and multipliers implementation as well as the basic principle of FIR filter.

5. In the Lab

The design steps you need to follow is shown below.



You are already well-conversant with Matlab and Modelsim from your last years and you have already used SystemVerilog for FIR filter design this year's Semester 1 lab. Feel free to use those resources.

A few things you need to remember here:

- A. The main purpose of Matlab analysis is to determine which filter is the best choice – FIR or IIR – for both noise and baseline wandering removal. This depends on the cut-off frequency and the order of the filter you will need to recover your clean ECG signal. Higher order filter means more hardware. You need extract the filter order and corresponding coefficients that will satisfy the required purpose of cleaning the ECG signal. Once you select your filter, you need to note down the filter coefficients – essentially you are extracting the h's (see FIR_DA.pptx in Resource folder as an example. The same concept applies for IIR filter although the design will be different).
- B. Direct multiplication is hardware and power intensive and therefore might not be a good solution unless the filter order is low – better to think in terms of either shift-and-add technique (see Multiplier.pptx in Resources) or Distributed Arithmetic (DA) to realise multiplierless multiplication.
- C. In principle, power consumption is proportional to the number of arithmetic operations and adders are least power hungry (depending upon what type of adder structure you use – see Adders.pptx).
- D. Given the above facts, it is not a good idea to write behavioural code in SystemVerilog. You should think more in terms of structural RTL design.
- E. Modelsim as such do not include timing – timing is a library-specific thing, i.e., an AND gate for example will show different timing characteristics in two different technology libraries as the technologies are different. Therefore, functional validation in Modelsim do not guarantee that your design will work in practice.
- F. You can check your timing only after synthesising on FPGA. During synthesis you need to put a timing constraint so that the synthesised circuit satisfy your timing requirement of 10 KHz. Remember, if the timing is not satisfied you might need to go over the design process once again to fix your architecture.
- G. As this exercise involves a bit of research element as well, I'll remain available for any discussion.

H. The Resources folder describes the design of FIR filter just as an example. In your work, depending on your Matlab analyses outcomes, you may decide to use an IIR filter structure. In that case, you can do a bit of research – there are tons of examples available for IIR filter structure in the internet – to implement your final HDL code.

6. Demonstration and report writing

The complete system needs to be demonstrated at 4.00 pm 9th March lab session. During the demonstration a short individual viva will be conducted. A report in the form of a research paper should be submitted via online system by 22/04/2026 4.00 pm. The report should clearly describe the entire work including performance analysis. The report will be 4-pages 2-column IEEE format with font size 10 Times New Roman.